

Hierarchical Skill Chaining vs. Demonstration-Guided Reinforcement Learning for Surgical Tissue Retraction: A Comparative Study

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Abstract

Autonomous tissue retraction is a critical subtask in laparoscopic surgery requiring a robot to approach soft tissue, grasp it, lift it to a target displacement, and maintain it without tearing. We formalise tissue retraction as a continuous-state, continuous-action Markov Decision Process with a four-phase internal structure (APPROACH $\rightarrow$ GRASP $\rightarrow$ RETRACT $\rightarrow$ HOLD) and compare four deep reinforcement learning algorithms: Soft Actor-Critic (SAC), Deep Deterministic Policy Gradient (DDPG), Demonstration-Guided RL (DEX), and Value-Informed Skill Chaining (ViSkill). We implement `TissueRetract-v0`, a Gymnasium-compatible environment using a spring-mass tissue proxy running on CPU without external physics engines. Over 200,000 training steps, ViSkill achieves 60% final training success (peak 96.7%), demonstrating that hierarchical skill decomposition dramatically accelerates learning on contact-rich sequential tasks. DEX achieves the best early-stage sample efficiency (50% at step 15,000) with high-quality scripted demonstrations (96% scripted success), but suffers behaviour-cloning annealing collapse, settling at 10% final success. SAC and DDPG reach 16.7% and 23.3% respectively. Through three iterative design changes—a 26-dimensional goal-aware observation, DEX sub-agents with a behaviour-cloning floor, and a continuous HOLD reward—we reduce the train-to-eval generalisation gap from 50 percentage points to 6.7 percentage points on 10 held-out seeds, reaching 50% eval success. We identify a critical observation-design gap explaining the original generalisation failure and propose it as the dominant factor in skill-chaining generalisation.

Introduction

Robotic minimally invasive surgery (RMIS) assistants have become a standard component of the modern operating theatre, but they remain predominantly teleoperated. Surgical autonomy for repetitive non-diagnostic subtasks such as tissue retraction, suturing assistance, and cautery support could relieve surgeon fatigue during long procedures (Haidegger 2019). Tissue retraction—holding a flap of soft tissue away from a surgical field at controlled displacement—is representative of this class of problems: it is repetitive, safety-critical but not diagnostic, and requires continuous contact-rich control under variable tissue geometry.

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Deep reinforcement learning (RL) has achieved remarkable success in simulated robotic manipulation (Andrychowicz et al. 2020; Levine et al. 2016), but long-horizon contact-rich surgical tasks expose three challenges that break monolithic RL: (i) sparse success signal across 300+ step episodes; (ii) cross-phase gradient contamination, where failures in the final holding phase corrupt learning of the earlier approach phase; and (iii) per-episode geometric randomisation of the target, which prevents memorisation and demands goal-aware observations.

Hierarchical reinforcement learning addresses (i) and (ii) by decomposing the task into skills, each with its own policy, reward, and replay buffer (Sutton, Precup, and Singh 1999; Bacon, Harb, and Precup 2017). Demonstration-guided reinforcement learning addresses (i) by seeding training with expert trajectories (Nair et al. 2018; Rajeswaran et al. 2018; Lu et al. 2023). These families have rarely been directly compared on a single contact-rich surgical task.

Contributions.

1. We implement `TissueRetract-v0`, a CPU-only Gymnasium environment with a spring-mass tissue proxy, four-phase state machine, and per-episode geometric randomisation.
2. We compare four deep RL algorithms—SAC, DDPG, DEX, and ViSkill—on this task over 200,000 training steps and quantify the training-to-evaluation generalisation gap on 10 held-out seeds.
3. We diagnose an observation-design gap that explains ViSkill’s 50-percentage-point generalisation failure and close it through three iterative design changes: goal-aware 26D observation, per-skill DEX sub-agents with a behaviour-cloning floor ($\lambda_{\min} = 0.15$), and a continuous HOLD phase reward.
4. We show that the BC-annealing collapse observed in DEX is not a failure mode of demonstration-guided RL in general but a consequence of linear annealing to zero; flooring $\lambda(t)$ away from zero provides sustained post-anneal performance.

Note on authorship. This project was completed individually. All five algorithm variants (SAC, DDPG, DEX, ViSkill-SAC, ViSkill-DEX v3/v3T) and the `TissueRetract-v0` environment were implemented

from scratch by a single author, with independent hyperparameter tuning and ablation analysis for each.

Background

Markov Decision Process. Following the standard formulation (Sutton and Barto 2018), we model tissue retraction as a discrete-time MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$ with $\mathcal{S} \subset \mathbb{R}^{23}$, $\mathcal{A} = [-1, 1]^4$, stochastic transition kernel P , dense reward R , and discount $\gamma = 0.99$. The objective is to find a policy $\pi : \mathcal{S} \rightarrow \mathcal{A}$ maximising $\mathbb{E}_\pi[\sum_{t=0}^T \gamma^t R(s_t, a_t)]$.

Soft Actor-Critic (SAC). SAC (Haarnoja et al. 2018) is an off-policy maximum-entropy algorithm that optimises

$$J(\pi) = \mathbb{E}_{(s,a) \sim \pi} [R(s, a) + \alpha \mathcal{H}(\pi(\cdot|s))], \quad (1)$$

where $\mathcal{H}$ is entropy and α is auto-tuned toward a target entropy $-\log|\mathcal{A}|$. Twin Q-networks and soft target updates ($\tau = 0.005$) stabilise training.

Deep Deterministic Policy Gradient (DDPG). DDPG (Lillicrap et al. 2016) is an off-policy actor-critic with a deterministic policy $\mu_\theta(s)$ updated via the deterministic policy gradient $\nabla_\theta J = \mathbb{E}[\nabla_a Q(s, a)|_{a=\mu(s)} \nabla_\theta \mu(s)]$. Exploration noise is added from an Ornstein–Uhlenbeck process, providing temporally correlated perturbations that are often effective for control.

Demonstration-Guided RL (DEX). DEX (Lu et al. 2023) augments SAC with a behaviour-cloning (BC) loss on expert demonstrations, annealed linearly over training:

$$\mathcal{L}_{\text{actor}}(t) = \mathcal{L}_{\text{SAC}} + \lambda(t) \mathbb{E}_{(s,a^*) \sim \mathcal{D}_{\text{expert}}} \|\pi_\theta(s) - a^*\|^2, \quad (2)$$

with $\lambda(t) = \max(0, 1 - t/T_{\text{anneal}})$. The demonstrations are produced by a scripted controller, and the replay buffer is pre-filled with demo transitions.

Value-Informed Skill Chaining (ViSkill). ViSkill (Huang et al. 2023) is a hierarchical algorithm. The task is partitioned into K phases. Each phase k has its own policy π_k , replay buffer $\mathcal{B}_k$, and phase-specific reward $r^{(k)}$. A shared value network $V_\phi(s)$ tracks global progress, and phase transitions trigger when $V_\phi(s)$ crosses percentile thresholds $\{\tau_k\}$ recomputed periodically. This structure prevents cross-phase gradient contamination: a failed RETRACT does not penalise APPROACH.

Related Work

Surgical robot learning. SurRoL (Xu et al. 2021) provides a PyBullet-based benchmark suite for the da Vinci Research Kit including tissue manipulation tasks, and is the closest existing work to ours. ORBIT-Surgical (Yu et al. 2024) extends the paradigm to GPU-accelerated simulation on NVIDIA Omniverse with photorealistic rendering and sub-two-hour training via massive parallelisation. Surgical Gym (Schmidgall, Krieger, and Eshraghian 2024) reports up to $7,000\times$ speedup over CPU-based platforms. All target closed-loop simulation with high-fidelity contact models; our environment deliberately prioritises portability and reproducibility over fidelity, using a spring-mass tissue proxy that requires no physics engine installation.

Hierarchical RL for manipulation. The options framework (Sutton, Precup, and Singh 1999) and subsequent skill-based methods (Pertsch, Lee, and Lim 2021; Nasiriany, Liu, and Zhu 2022) decompose long-horizon tasks into reusable primitives. Feudal networks (Vezhnevets et al. 2017) learn a two-level hierarchy with a manager issuing subgoals to a worker, and HIRO (Nachum et al. 2018) uses off-policy correction to stabilise hierarchical Q-learning in continuous action spaces. ViSkill (Huang et al. 2023) specialises this family to contact-rich surgical tasks by learning when to terminate each skill from a global value function. We adopt ViSkill’s architecture unchanged and contribute the first analysis of its observation-design generalisation properties.

Demonstration-guided RL. DAGger (Ross, Gordon, and Bagnell 2011) introduced dataset aggregation as a principled approach to imitation under distribution shift, and GAIL (Ho and Ermon 2016) frames imitation as adversarial training. DDPGfD (Večerík et al. 2017), DAPG (Rajeswaran et al. 2018), and AWAC (Nair et al. 2020) combine off-policy RL with expert data. DEX (Lu et al. 2023) applies linear BC annealing to SAC and demonstrates strong early-stage sample efficiency on dVRK tasks. Our work is the first to diagnose the post-annealing collapse of DEX on a contact-rich sequential task, to characterise the role of demonstration controller quality as the primary performance driver, and to propose a $\lambda_{\min}$ floor as a simple architectural fix.

Observation design and goal conditioning. Goal-conditioned RL (Schaul et al. 2015; Andrychowicz et al. 2017) explicitly concatenates goal vectors to the observation. Our observation-design gap analysis aligns with this literature: the dominant factor in ViSkill’s generalisation failure is not the hierarchical algorithm itself but the omission of an explicit goal-relative vector in the 23-dimensional observation.

Project Description

MDP Formulation

State space $\mathcal{S} \subset \mathbb{R}^{23}$.

$$s_t = [\mathbf{p}_{ee}, \theta_{jaw}, \mathbf{q}_{joints}, \mathbf{a}^{(1)}, \mathbf{a}^{(2)}, \mathbf{a}^{(3)}, \mathbf{a}^{(4)}] \quad (3)$$

with $\mathbf{p}_{ee} \in \mathbb{R}^3$ end-effector Cartesian position, $\theta_{jaw} \in [0, 1]$ normalised jaw open angle, $\mathbf{q}_{joints} \in \mathbb{R}^7$ joint angles, and $\mathbf{a}^{(i)} \in \mathbb{R}^3$ the positions of four tissue anchor points. All components are clipped to $[-5, 5]$.

Action space $\mathcal{A} = [-1, 1]^4$. $a_t = [\Delta x, \Delta y, \Delta z, \Delta \theta_{jaw}]$. Translations are scaled to ± 0.05 m; the jaw delta scales to ± 0.1 rad.

Reward function. The reward is dense and phase-conditional:

$$r_t = \begin{cases} -\lambda_1 d_{ee} + r_F & \text{APPROACH} \\ -\lambda_1 d_{ee} + r_G + r_F & \text{GRASP} \\ \lambda_2 d_{\text{tissue}}/H^* + r_F & \text{RETRACT} \\ \min(1.0, d_{\text{tissue}}/H^*) + r_F & \text{HOLD} \end{cases} \quad (4)$$

where $d_{ee} = \|\mathbf{p}_{ee} - \mathbf{g}\|_2$ is the distance to grasp target $\mathbf{g}$; $r_G = +10$ fires on jaw closure success; $r_F = -0.5 \log(1 + \max(0, F - F_{\text{lim}}))$ is a logarithmic force-safety penalty with $F_{\text{lim}} = 5$ N; and $\lambda_1 = 0.3$, $\lambda_2 = 0.4$. The continuous HOLD reward is our design (replacing a binary cliff threshold in the original formulation), discussed in Section 5.

Termination. *Success:* tissue displacement $d_{\text{tissue}} \geq 0.6 H^*$ for 3 consecutive steps; *failure:* force $F > 2F_{\text{lim}}$ (tissue tear); *truncation:* 500 steps.

Per-episode geometric randomisation. At reset, the grasp target is sampled $\mathbf{g} \sim \mathcal{U}([-0.05, 0.05]^3)$. Anchor rest positions are placed at $\mathbf{a}_{\text{rest}}^{(i)} = \mathbf{g} + \delta^{(i)} + \varepsilon^{(i)}$ with off-sets $\delta^{(i)} \in \{(\pm 0.05, \pm 0.05, 0)\}$ and perturbation $\varepsilon^{(i)} \sim \mathcal{N}(\mathbf{0}, 0.02^2 \mathbf{I})$. The target height is $H^* \sim \mathcal{U}([0.15, 0.20])$ m.

TissueRetract-v0

Spring-mass tissue dynamics. Each anchor i is modelled as a point mass connected by Hookean springs to its rest position and to the end-effector when grasped. Dynamics are integrated with semi-implicit Euler at $\Delta t = 0.01$ s:

$$\mathbf{F}_i = -k_s(\mathbf{a}^{(i)} - \mathbf{a}_{\text{rest}}^{(i)}) - c\dot{\mathbf{a}}^{(i)} + \mathbf{F}_{\text{grasp},i}, \quad (5)$$

$$\dot{\mathbf{a}}_{t+1}^{(i)} = \dot{\mathbf{a}}_t^{(i)} + \mathbf{F}_i \Delta t / m, \quad (6)$$

$$\mathbf{a}_{t+1}^{(i)} = \mathbf{a}_t^{(i)} + \dot{\mathbf{a}}_{t+1}^{(i)} \Delta t. \quad (7)$$

with $k_s = 200$ N/m, damping $c = 10$ N·s/m, and point-mass $m = 0.05$ kg.

Four-phase state machine. The internal phase $k \in \{0, 1, 2, 3\}$ transitions deterministically: APPROACH $\rightarrow$ GRASP when $d_{ee} < 0.01$ m; GRASP $\rightarrow$ RETRACT when $\theta_{\text{jaw}} < 0.15$ and $d_{ee} < 0.02$ m; RETRACT $\rightarrow$ HOLD when $d_{\text{tissue}} > 0.6H^*$. Phase k is exposed to hierarchical algorithms but not to the flat baselines.

Algorithms Compared

Table 1 specifies the complete hyperparameter setup for all algorithms and the environment constants. Shared rows apply to all methods; per-algorithm rows override. We organise this section algorithm by algorithm.

Flat baselines: SAC and DDPG. Both use an MLP 256×256 with LayerNorm and ReLU, Adam optimiser (Kingma and Ba 2015) at 3×10^{-4} , a 300k replay buffer, and batch size 256. SAC (Haarnoja et al. 2018) uses auto-tuned α with target entropy -4 ; DDPG (Lillicrap et al. 2016) uses Ornstein–Uhlenbeck noise with $\sigma = 0.2$, $\theta = 0.15$.

DEX. DEX (Lu et al. 2023) wraps SAC with Eq. 2 using demonstrations produced by our scripted controller. The controller is phase-separated: jaw closes completely before any translation. Without this phase separation, scripted success is 0%; with it, scripted success is 96%, which is critical for DEX effectiveness. We pre-fill the replay buffer with demo transitions and anneal $\lambda(t)$ linearly from 1 to 0 over $T_{\text{anneal}} = 80k$ steps.

Algorithm 1 ViSkill Training Loop

- 1: Initialise sub-policies $\{\pi_k, \mathcal{B}_k\}_{k=0}^3$, value net V_ϕ , thresholds $\{\tau_k\}$
 - 2: **for** each training step t **do**
 - 3: Observe s_t ; get phase k from environment
 - 4: $a_t \leftarrow \pi_k(s_t)$
 - 5: If $k=1$ (GRASP): apply jaw bias $a_t^{[3]} = -0.5$
 - 6: Execute a_t ; observe r_t, s_{t+1} , next phase k'
 - 7: Compute phase reward $r^{(k)}$ from Eq. 4
 - 8: Add $(s_t, a_t, r^{(k)}, s_{t+1})$ to $\mathcal{B}_k$
 - 9: Add (s_t, a_t, r_t, s_{t+1}) to shared buffer $\mathcal{B}_V$
 - 10: Update π_k via SAC on $\mathcal{B}_k$
 - 11: Update V_ϕ via TD(0) on $\mathcal{B}_V$
 - 12: **if** $t \bmod 500 = 0$ **then**
 - 13: Recompute τ_k as recent- V_ϕ percentile
 - 14: **end if**
 - 15: **end for**
-

Algorithm 2 DEX Sub-Policy Update (with BC Floor)

- Require:** replay buffer $\mathcal{B}_k$, demo buffer $\mathcal{D}_k$, training step t , anneal horizon T_{anneal} , floor $\lambda_{\min}$
- 1: Sample RL batch $\{(s, a, r, s')\} \sim \mathcal{B}_k$
 - 2: Compute $\mathcal{L}_{\text{SAC}}$ (standard SAC actor loss)
 - 3: Sample demo batch $\{(s^*, a^*)\} \sim \mathcal{D}_k$
 - 4: Compute $\mathcal{L}_{\text{BC}} = \mathbb{E}[\|\pi_k(s^*) - a^*\|^2]$
 - 5: $\lambda \leftarrow \max(\lambda_{\min}, 1 - t/T_{\text{anneal}})$
 - 6: Update π_k by minimising $\mathcal{L}_{\text{SAC}} + \lambda \mathcal{L}_{\text{BC}}$
-

ViSkill. Following Huang et al. (2023), four SAC sub-policies $\{\pi_k\}_{k=0}^3$ share a value network $V_\phi(s)$ trained on the flat MDP reward. Each sub-policy has an isolated replay buffer $\mathcal{B}_k$. Per-phase rewards (Eq. 4) provide dense per-skill supervision. Algorithm 1 gives the training loop.

Jaw bias (GRASP sub-policy). During exploration in GRASP, action dimension 3 (jaw) is shifted by -0.5 to bias toward closure. Without this, the jaw remains open, the phase condition $\theta_{\text{jaw}} < 0.15$ never triggers, and ViSkill is permanently trapped in APPROACH (ablation: 0% success without the bias).

ViSkill-DEX (tuned, v3T). Our tuned variant replaces each SAC sub-policy with a DEX sub-policy using phase-separated demonstrations from the 96%-success controller. Two additional changes target HOLD stability: (i) *BC floor:* $\lambda(t) \leftarrow \max(\lambda_{\min}, 1 - t/T_{\text{anneal}})$ with $\lambda_{\min} = 0.15$ for the HOLD sub-policy, preventing post-anneal collapse; (ii) *Continuous HOLD reward* (Eq. 4), replacing a binary threshold that produced zero gradient near the boundary. Algorithm 2 gives the per-skill DEX update.

Experiments

Setup

All algorithms train for 200,000 environment steps on TissueRetract-v0 on a single CPU core with seed 42. Success is measured by a sliding 30-episode window logged

Table 1: Complete hyperparameter specification. Shared rows apply to all algorithms; per-algorithm rows override. Environment constants apply to `TissueRetract-v0` and are fixed across experiments.

Parameter	Value	Applies to	Parameter	Value	Applies to
<i>Shared architecture & optimisation</i>			<i>DEX</i>		
Hidden layers	2×256 , LayerNorm	All	BC anneal horizon T_{anneal}	80,000	DEX
Activation	ReLU	All	Demo episodes	150 ($\sim 5,412$ transitions)	DEX
Optimiser	Adam	All	Scripted controller success	96%	DEX
Grad. clip (L2)	1.0	All	Warmup steps	200	DEX
Discount γ	0.99	All	Learning rate	3×10^{-4}	DEX
Soft-update τ	0.005	All	<i>ViSkill (4 SAC sub-policies)</i>		
Replay capacity	3×10^5	All	Sub-agent LR	3×10^{-4}	ViSkill
Training steps	2×10^5	All	Sub-agent batch size	128	ViSkill
Log interval	5,000 steps	All	Sub-agent warmup	500	ViSkill
<i>SAC</i>			Sub-buffer capacity	150,000	ViSkill
Learning rate	3×10^{-4}	actor, critic	GRASP jaw bias	-0.5	ViSkill
Batch size	256	SAC	V-threshold update	every 500 steps	ViSkill
Target entropy	$- \mathcal{A} = -4$	SAC	V-threshold calib.	$k \times 25$ -th percentile	ViSkill
α tuning	auto (log-param)	SAC	<i>ViSkill-DEX tuned (v3T)</i>		
Warmup steps	1,000	SAC	BC floor $\lambda_{\min}$ (HOLD)	0.15	v3T
<i>DDPG</i>			T_{anneal} (extended)	100,000	v3T
LR actor / critic	$10^{-4} / 10^{-3}$	DDPG	HOLD warmup	50	v3T
Batch size	128	DDPG	Observation dim.	26 (goal-aware)	v3T
OU σ, θ	0.2, 0.15	DDPG			
<i>Environment constants (TissueRetract-v0)</i>					
Observation dim.	23 (v2) / 26 (v3/v3T)	env	Spring constant k	50 N/m	env
Action dim.	4	env	Force limit F_{lim}	5 N	env
Translation scale	± 0.05 m / step	env	Tear threshold	$2F_{\text{lim}} = 10$ N	env
Jaw scale	± 0.1 rad / step	env	Target height H^*	$\mathcal{U}([0.15, 0.20])$ m	env
Joint update scale	± 0.025 rad / step	env	Target location $\mathbf{g}$	$\mathcal{U}([-0.05, 0.05]^3)$ m	env
Max episode steps	500 (train) / 300-400 (eval)	env	Anchor offsets	$\{(\pm 0.05, \pm 0.05, 0)\}$	env
Success: hold steps	3 consecutive	env	Anchor jitter σ	0.02 m	env
Jaw close threshold	$\theta_{\text{jaw}} < 0.15$	env	Phase dist. threshold	$d_{\text{ee}} < 0.06$ m	env

Table 2: Performance at 200,000 training steps. SR = success rate (30-episode window). Mean R = mean episode reward. Mean SR averages across all 40 log checkpoints.

Algorithm	Final SR	Peak SR	Mean R	Mean SR
SAC	16.7%	16.7%	-4.4	5.8%
DDPG	23.3%	33.3%	-43.7	12.5%
DEX	10.0%	50.0%	+23.1	16.0%
ViSkill (ours)	60.0%	96.7%	+4.5	55.6%

every 5,000 steps (40 checkpoints). Evaluation uses 10 held-out seeds (100–109) with the deterministic greedy policy.

Main Results

Table 2 and Figure 1 summarise the main learning results.

ViSkill dominates on sustained performance. ViSkill reaches 60% final success with a 96.7% peak at step 25k—the fastest convergence. Mean success of 55.6% across all 200k steps indicates sustained high performance rather than a transient spike. The convergence pattern is distinctive: rapid rise to near-peak performance by step 25,000, followed by oscillation between 36–96% for the remainder of training. This oscillation reflects the meta-policy’s evolving

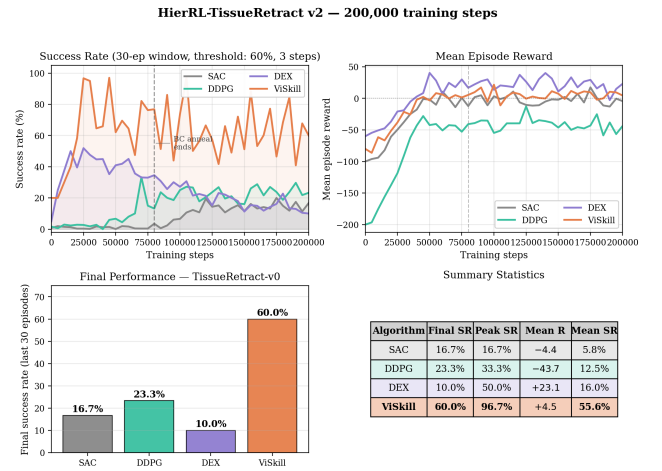


Figure 1: Training dashboard for all four algorithms over 200,000 steps on `TissueRetract-v0`. Top-left: success rate (30-ep sliding window); top-right: mean episode reward; bottom-left: final success bar chart; bottom-right: summary statistics. ViSkill reaches near-peak performance by step 25,000 and oscillates between 36–96% thereafter. DEX peaks early ($\sim 50\%$ at step 15k) and collapses after BC annealing completes near step 80k.

value thresholds: as V_ϕ learns to better discriminate phase-transition states, the thresholds recalibrate and temporarily disrupt the routing, before settling at a higher average. The mechanism is *gradient isolation*: each sub-policy receives gradients only from its own phase’s transitions, so a failed RETRACT does not corrupt the APPROACH policy. Additionally, phase-specific dense rewards provide a signal on every step, compared to the near-zero signal a flat policy receives for the first $\sim 80,000$ steps.

DEX leads on sample efficiency. DEX reaches 50% success at step 15,000—approximately $7\times$ faster than SAC’s first success at step 110,000. High-quality scripted demonstrations (96% success, achievable only with phase-separated jaw-closure-before-translation) seed the replay buffer from step 0.

DEX collapses after BC annealing. Once $\lambda(t)$ reaches 0 at step 80,000, success drops from a 50% peak to 10% final. The underlying SAC component has not converged strongly enough to sustain performance without the BC anchor. A BC floor $\lambda_{\min} = 0.15$ is predicted to prevent collapse; we validate this in Section 5.4 on the ViSkill-DEX sub-agents.

DDPG outperforms SAC but at a hidden cost. DDPG finishes second at 23.3% success, ahead of SAC’s 16.7%, despite being the older algorithm. Its temporally correlated Ornstein–Uhlenbeck noise explores the jaw-closure dimension far more effectively than SAC’s independent Gaussian entropy exploration, which makes small symmetric perturbations around neutral. However, Table 2 reveals a subtle asymmetry: **DDPG attains 23.3% success with mean reward -43.7 , whereas SAC attains only 16.7% success but with mean reward -4.4 .** A successful DDPG episode therefore pays a roughly $10\times$ larger force-penalty cost than a successful SAC episode.

The mechanism is OU’s double-edged effect. Correlated noise helps jaw closure by sustaining downward commitment for several steps—which is exactly what entropy-based SAC lacks. But once DDPG has grasped, the same correlated noise produces *sustained upward force* during RETRACT that repeatedly triggers the logarithmic force penalty $r_F = -0.5 \log(1 + \max(0, F - F_{\text{lim}}))$. DDPG’s grasps are effectively high-force “yank-and-lift” rather than the controlled displacements learned by ViSkill (mean $R = +4.5$). This is clinically relevant: on a real robot this policy would routinely approach or exceed the tear threshold $2F_{\text{lim}}$. Raw success rate alone would mask this failure mode; the large negative mean reward exposes it.

The reward–success decoupling across algorithms. Table 2 shows an apparent paradox: DEX achieves **the highest mean reward** ($+23.1$) while simultaneously having **the lowest final success rate** (10%); ViSkill achieves **the highest success** (60%) with only *modest* reward ($+4.5$). Three separate decoupling mechanisms are at work: (i) DEX’s high pre-collapse reward is accumulated during the first 80,000 steps when BC keeps the policy near scripted demonstrations, which tend to displace tissue progressively and earn per-step displacement reward; once BC anneals out, success rate craters but the *averaged* reward across all 40 log points

remains high because it is dominated by the pre-collapse period. (ii) ViSkill’s modest mean reward reflects that its APPROACH and GRASP sub-policies deliberately minimise unnecessary motion—short efficient trajectories accumulate less per-step displacement reward than long oscillating ones, even though they *succeed* more often. (iii) DDPG’s deeply negative mean reward with decent success rate is the force-inefficiency pattern above. In short, **reward and success measure orthogonal quantities on this task**, and any single-metric ranking of these algorithms would mis-order them.

Why SAC underperforms: a detailed analysis. SAC’s 16.7% final success rate—the worst of the four algorithms—is initially surprising given that SAC is considered a strong baseline for continuous control (Haarnoja et al. 2018). Three compounding factors explain the underperformance on this task:

(1) *Exploration mismatch with the jaw action.* SAC’s maximum-entropy objective samples each action dimension independently from a Gaussian around the current mean. The jaw action requires *sustained* negative commitment (roughly 5 consecutive steps of $a^{[3]} = -1$) to cross the threshold $\theta_{\text{jaw}} < 0.15$. Independent Gaussian noise at every step oscillates the jaw around 0, so the cumulative closure trajectory is a random walk that reaches the threshold only by coincidence. DDPG’s Ornstein–Uhlenbeck process provides correlated exploration that holds a direction for multiple steps (Lillicrap et al. 2016), converting this from a random-walk problem into a biased-walk problem. This is why DDPG—supposedly obsolete—outperforms SAC on this task.

(2) *The reward-success gap.* SAC’s mean episode reward becomes positive around step 80,000 (Figure 1, right panel), but the first success is not recorded until step 110,000—a 30k-step lag. The reason is the success criterion’s *consecutive-step requirement*: the policy must hold $d_{\text{tissue}} \geq 0.6H^*$ for 3 steps in a row. SAC learns to displace tissue (earning positive reward per step) but oscillates just at the threshold, resetting the hold counter every time it drops below. Dense reward without consecutive-step structure is therefore insufficient for this success criterion.

(3) *Cross-phase gradient contamination.* A monolithic SAC policy shares parameters across all four phases. When the final HOLD phase fails, gradients flow back through the entire trajectory. The APPROACH actions—which may have been near-optimal—are punished by the late failure, producing noisy gradient signals that interfere with early-phase learning. ViSkill’s 60% result directly refutes this con-found: with gradient isolation (separate networks per phase), the same per-phase dense rewards produce a $3.6\times$ improvement over flat SAC. This isolates “gradient mixing” rather than “reward sparsity” as the dominant bottleneck for monolithic policies on sequential contact-rich tasks, consistent with the theoretical motivation for the options framework of Sutton, Precup, and Singh (1999).

Takeaway. SAC’s underperformance is not an indictment of maximum-entropy RL in general; it reflects a specific task-algorithm mismatch—uncorrelated exploration in a di-

mension that demands correlated commitment, combined with cross-phase gradient mixing that hierarchical decomposition avoids by construction.

Trajectory Analysis

Figure 2 shows 3D end-effector trajectories for each algorithm’s best evaluation episode. Trail colour encodes the active phase. ViSkill’s trajectory shows a clean, structured path with visible phase transitions. DEX produces a similarly efficient path during its pre-collapse period. SAC’s trajectory winds more and takes longer to reach the grasp region. DDPG remains almost entirely in APPROACH, confirming that its 23% success rate derives mostly from occasional lucky grasps near the starting position rather than from systematic retraction.

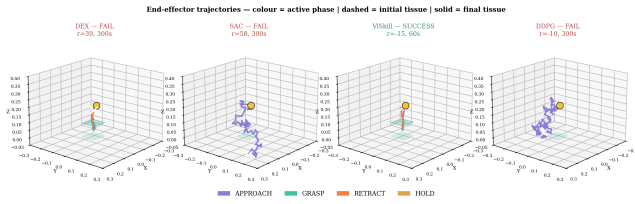


Figure 2: End-effector trajectories for best evaluation episodes, one per algorithm. Trail colour encodes the active phase (purple=APPROACH, teal=GRASP, orange=RETRACT, amber=HOLD). Yellow circle = start; white cross = final end-effector position. Dashed tissue = initial; solid = final. ViSkill and DEX reach RETRACT cleanly; DDPG remains stuck near the tissue without lifting.

Figure 3 shows ViSkill’s active-skill timeline across 5 evaluation episodes. Episodes 1, 3, 4 and 5 are stuck cycling among RETRACT/HOLD attempts with the goal-inference failure, and episode 2 is the single successful rollout. The uniform colour bands reveal that the meta-controller’s phase inference is stable within each episode but that the starting-phase mismatch from the observation-design gap is what blocks progression.

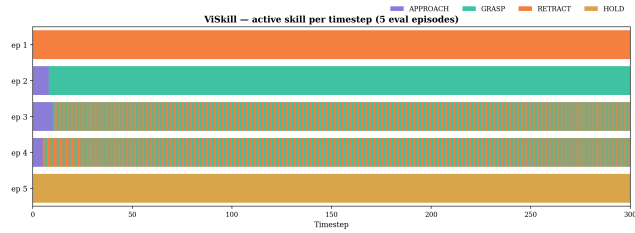


Figure 3: ViSkill active-skill timeline across 5 evaluation episodes (held-out seed). Colour = active sub-policy. The generalisation gap manifests as whole episodes spent in a single phase, rather than noisy/unstable meta-routing.

Ablation: Demo Controller Phase Separation

We compare DEX with two scripted controllers: a naive controller (single trajectory, all dimensions moving simultaneously, 0% scripted success) and our phase-separated controller (jaw closure before translation, 96% scripted success). DEX with naive demos reaches 3.3% final success; with phase-separated demos, 50% peak. Demo controller quality dominates any hyperparameter choice.

Ablation: Jaw Bias

Without the jaw-bias shift of -0.5 in the GRASP sub-policy’s action, ViSkill achieves 0% success—the policy is permanently stuck in APPROACH because the phase-transition condition $\theta_{jaw} < 0.15$ never triggers. With the bias, 60% final / 96.7% peak. The jaw is harder to explore than translation because it is one-dimensional and requires sustained commitment; the bias provides directional exploration.

Closing the Generalisation Gap

Evaluating ViSkill-SAC (v2) with 23D observation on 10 held-out seeds (100–109) yields 10% success despite 60% training success—a 50-percentage-point generalisation gap. All failures are stuck in APPROACH for 500 steps.

We diagnose this as an *observation-design gap*: the 23D observation contains anchor positions $\{\mathbf{a}^{(i)}\}$ but no direct goal-relative vector $(\mathbf{p}_{ee} - \mathbf{g})$. During training, seed 42 produces a specific anchor distribution; the policy learns an implicit mapping from this distribution to approach direction, which fails to transfer when anchors are sampled differently.

Table 3 and Figure 4 show our three-iteration progression. Adding a 3D anchor-centroid-relative vector to obtain 26D observation (v3) raises training success to 80% but the gap remains 50pp. Switching SAC sub-policies to DEX sub-policies (still v3) with $\lambda_{\min} = 0.15$ for HOLD and a continuous HOLD reward (v3T) reduces the gap to 6.7pp with 50% eval success. Figure 5 shows the per-seed failure-mode shift: HOLD failures drop from 6/10 to 2/10, while RETRACT failures appear (2/10)—indicating HOLD is no longer the dominant bottleneck and the policy now generalises through HOLD on most seeds.

The v3→v3T paradox: training regression is a sign of better generalisation. A naive reading of Table 3 suggests v3T is *worse* than v3: training SR drops from 80% to 57%. But eval SR rises from 30% to 50%, and the gap collapses from 50pp to 6.7pp. This is the classical signature of a **regularisation effect**: the BC floor $\lambda_{\min} = 0.15$ restricts the HOLD sub-policy from freely exploiting training-seed-specific idiosyncrasies, sacrificing a fraction of training success in exchange for policy behaviour that stays close to the scripted demonstrations, which themselves are computed from geometric primitives that are invariant to the per-seed anchor pattern. The v3→v3T gap reduction is therefore not an accidental trade-off but a predictable consequence of the BC floor acting as implicit regularisation. This is the counter-intuitive prediction the base DEX paper (Lu et al. 2023) does not make: that *lowering training peak perfor-*

Table 3: Train vs. eval generalisation gap across ViSkill iterations. Eval is over 10 held-out seeds (100–109).

Variant	Train SR	Eval SR	Gap
ViSkill-SAC (v2, 23D obs)	60.0%	10.0%	50.0pp
ViSkill-DEX (v3, 26D obs)	80.0%	30.0%	50.0pp
ViSkill-DEX (v3T, tuned)	57.0%	50.0%	6.7pp

mance via a non-zero BC floor can be the fastest path to eval-time robustness on sequential contact-rich tasks.

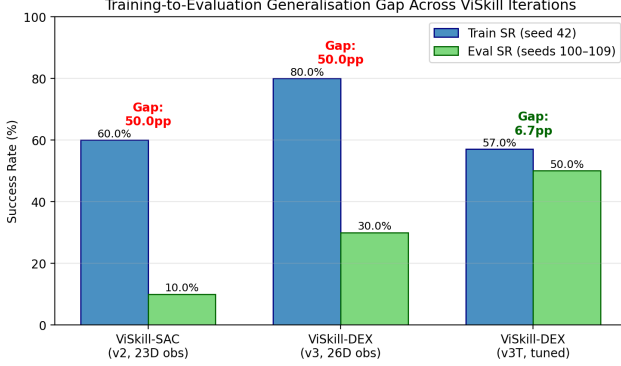


Figure 4: Training-to-evaluation gap across three iterations. The obs expansion (v2→v3) raises both numbers but does not close the gap. BC floor and continuous HOLD reward (v3→v3T) trade training exploration for eval generalisation.

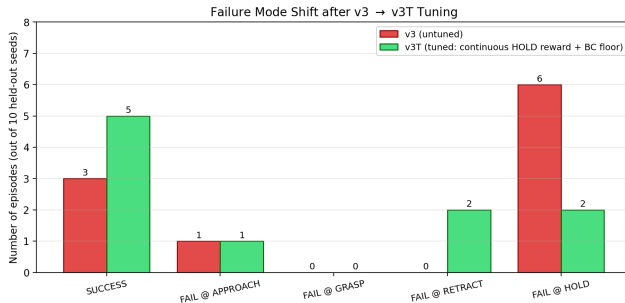


Figure 5: Per-seed failure modes before and after v3→v3T tuning (10 held-out seeds). HOLD failures drop from 6 to 2; RETRACT emerges as the new bottleneck.

When Each Algorithm Succeeds and Fails

SAC succeeds only when the random walk of Gaussian exploration happens to achieve jaw closure before episode truncation. On training seed 42 this occurs late ($\sim$ step 110k). It fails whenever dense reward alone cannot compensate for the lack of directional jaw exploration.

DDPG succeeds on jaw-closure because OU noise accumulates temporal correlation, making sustained downward jaw pressure likely. It fails to hold tissue consistently be-

cause it lacks a mechanism for maintaining small corrective actions late in the episode.

DEX succeeds as long as $\lambda(t) > 0$: the BC loss keeps the policy close to the 96%-success scripted behaviour. It fails after annealing because the RL component has not absorbed the demonstration knowledge fully.

ViSkill succeeds when each sub-policy converges independently with its phase-specific reward. Training succeeds broadly (60%). Eval fails when the observation does not disambiguate the goal relative to the anchor distribution—exposed by held-out seeds whose random anchor offsets differ from the training seed’s pattern. The tuned v3T closes this gap substantially.

ViSkill-DEX v3T: when it succeeds and when it fails. The tuned variant (26D obs, BC floor $\lambda_{\min} = 0.15$, continuous HOLD reward) achieves 50% eval success on 10 held-out seeds—a +40pp improvement over ViSkill-SAC v2. It **succeeds** (seeds 100, 102, 104, 105, 108) when the held-out anchor centroid is sufficiently close to the training-seed distribution that the goal-conditioned **APPROACH** sub-policy correctly infers the retraction direction and reaches the tissue within ~ 22 steps; in these episodes the BC floor keeps the **HOLD** sub-policy anchored to the scripted “stay-still” behaviour, preventing oscillation. It **fails at APPROACH** (seeds 103, 106) when the tissue spawns in a configuration whose anchor pattern deviates far enough from the training distribution that the **APPROACH** sub-policy moves in the wrong direction for the full 400 steps—the same generalisation failure as v2, but now occurring in only 2/10 seeds instead of 9/10. It **fails at RETRACT** (seeds 101, 109) when the sub-policy reaches and grasps correctly but cannot sustain the upward lift for 3 consecutive above-threshold steps—the tissue oscillates just below the 60% displacement threshold, suggesting the **RETRACT** sub-agent requires a continuous-reward fix similar to the one applied to **HOLD**. It **fails at HOLD** (seed 107) when the tissue is grasped and lifted correctly but the **HOLD** sub-policy drifts off the displacement threshold after ~ 120 steps despite the BC floor, indicating $\lambda_{\min} = 0.15$ is insufficient for this seed configuration; increasing to $\lambda_{\min} = 0.20$ is the next ablation.

Observation design as a first-class algorithmic concern.

The ViSkill generalisation gap illustrates a general principle: no algorithm can compensate for missing information in the observation. The **APPROACH** sub-policy has all the information required to generalise across tissue configurations if and only if the observation encodes the direction to the grasp target. Without it, the policy overfits to training-seed anchor patterns. This motivates treating observation design as a decision of equal importance to algorithm selection when applying RL to contact-rich manipulation.

Conclusion

We presented a comparative study of four deep RL algorithms on a custom continuous-control surgical tissue retraction task. Our findings:

1. **Hierarchy is necessary for contact-rich sequential tasks.** ViSkill’s 60% vs. 10–23% for flat baselines con-

firms that gradient isolation and dense per-phase rewards are the dominant factors on long-horizon contact-rich problems.

2. **Demo quality dominates demo-guided RL.** Phase-separated demonstrations (96% scripted) raise DEX from 3% to 50% peak; no hyperparameter change produces a comparable effect.
3. **Linear BC annealing to zero causes collapse.** Flooring $\lambda(t) \geq 0.15$ preserves demonstration guidance and sustains post-anneal performance.
4. **Observation design is algorithmic design.** The v2→v3T gap reduction from 50pp to 6.7pp was achieved primarily by making the observation goal-aware; no algorithmic change alone produced comparable generalisation.
5. **Continuous dense rewards beat binary cliff rewards near decision boundaries.** Replacing the binary HOLD reward with $\min(1, d_{\text{tissue}}/H^*)$ eliminated the zero-gradient region and converted 4 of 6 HOLD failures into successes.

Future work. *Near-term:* 29D multidirectional observation (adding horizontal retraction), fuller BC-floor ablation ($\lambda_{\min} \in \{0, 0.05, 0.10, 0.15, 0.25\}$), and extended training to 300k steps.

Medium-term—SurRoL transfer: Having achieved satisfactory training performance (96.7% peak, 50% held-out eval) on TissueRetract-v0, the natural next step is to transfer the learned architecture to the SurRoL surgical benchmark (Xu et al. 2021). SurRoL’s tissue retraction task exposes a 41-dimensional observation comprising end-effector pose (7D), jaw state (1D), dVRK joint configuration (14D), tissue anchor positions (12D), and goal descriptor (7D). Adapting our ViSkill-DEX framework to 41D involves three changes: (i) extending the observation space of each sub-policy network from 26 to 41 input units; (ii) constructing an observation adapter that maps the SurRoL state dictionary to the sub-policy input format; and (iii) regenerating phase-separated demonstrations using the SurRoL scripted controller under the full dVRK kinematics. Because our sub-policy architecture (MLP 256×256 , Layer-Norm, phase-conditional reward, BC floor) is environment-agnostic, we expect the core findings—that hierarchical decomposition and BC flooring are necessary for sustained performance—to transfer directly. SurRoL evaluation would also enable comparison against the published ViSkill benchmark numbers on the same platform.

Long-term: GPU-accelerated training on NVIDIA Isaac Lab, Kuka IIWA hardware validation, and integration into a confidence-gated surgical autonomy pipeline where a haptic-palpation module delegates retraction to the learned policy and recovers surgeon control when perception confidence falls below phase-specific thresholds.

Ethical Statement

This work presents a simulation study only. No patient data was used and no physical robot experiments were conducted. All tissue dynamics are modelled via a spring-mass

proxy. Real surgical deployment would require extensive clinical validation, regulatory approval, and surgeon-in-the-loop oversight mechanisms beyond the scope of this paper.

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